

ZYVORAI LABS · CUSTOMER DOCUMENTATION

cloud-netconfig

Getting Started

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cloud-netconfig applies secondary IPs and policy routing from Azure, AWS EC2, and GCP instance metadata — event-driven via netlink, so changes propagate without manual route edits. This page gets you from install to a running daemon.

Prerequisites

- Root/sudo access on the host (installs a systemd unit and modifies routes).
- The host must run on a supported cloud (Azure, AWS EC2, or GCP) so instance metadata is reachable.
- Rust toolchain and `make` to build from source (see the upstream README for exact versions).

1. Build and install

```
git clone https://github.com/hypersdk/cloud-netconfig.git
cd cloud-netconfig
make build
sudo make install
sudo useradd -M -s /usr/bin/nologin cloud-network 2>/dev/null || true
```

This builds and installs `cloud-netconfigd` (daemon) and `cnctl` (CLI).

2. Activate configuration

Packaging installs the annotated example as `/etc/cloud-network/config.yaml` . The daemon reads `/etc/cloud-network/cloud-network.yaml` — copy or symlink before first start:

```
sudo cp /etc/cloud-network/config.yaml /etc/cloud-network/cloud-network.yaml  
or: sudo ln -sf /etc/cloud-network/config.yaml /etc/cloud-network/cloud-network.yaml
```

Edit supplementary interfaces (`network.interfaces.enabled`) for your VM's secondary NICs.

3. Start the daemon

```
sudo systemctl enable --now cloud-netconfd
```

4. Verify

```
cnctl status all  
curl -s http://127.0.0.1:5209/health  
curl -s http://127.0.0.1:5209/api/status
```

`cnctl status all` prints daemon status, detected cloud provider, and local interface addresses. The local HTTP API defaults to `127.0.0.1:5209`.

Troubleshooting

- `cnctl status` shows "**Daemon Status: not running**" — confirm `systemctl status cloud-netconfigd` and that port `5209` is listening on localhost.
- **Daemon exits immediately** — cloud auto-detection failed; the host must be on Azure, AWS, or GCP. Check `journalctl -u cloud-netconfigd -n 50`.
- **No secondary routes** — confirm `network.interfaces.enabled` lists the correct NIC names and metadata assigns those interfaces.

Next steps

- [Configuration](#)
- [Common workflows](#)
- [Admin basics](#)

Operate (CLI)

1. Confirm the unit is active: `systemctl is-active cloud-netconfigd` .
 2. Read daemon health: `curl -s http://127.0.0.1:5209/api/status` .
 3. Inspect host networking: `cnctl status network` .
 4. **Empty / fail:** Check `journalctl -u cloud-netconfigd -f` , cloud metadata reachability, and config path (`cloud-network.yaml`).
 5. **Success:** Status shows `running` , a provider name, and configured interfaces with addresses.
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cloud-netconfig is operated through **cnctl** and the **cloud-netconfigd** systemd service — there is no first-party web console.

Essentials

```
ncctl status api
ncctl status network
ncctl apply --dry-run
ncctl version
curl -s http://127.0.0.1:5209/api/status
```

Where to go next

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First install	Getting Started
Config format	Configuration
Day-2 jobs	Workflows
Host / packaging	Admin Basics
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Operate (CLI)

1. Run `cnctl status all` after any install or config change.
2. Use `cnctl apply --dry-run` before copying YAML to `/etc/cloud-network/cloud-network.yaml`.
3. **Empty / fail:** Fall back to `journalctl -u cloud-netconfigd` and the `/health` endpoint.
4. **Success:** Commands exit 0 and status output matches the cloud VM's metadata and interfaces.

Workflow	Steps
Fresh install	Clone → <code>make build</code> → <code>sudo make install</code> → symlink/copy config → <code>systemctl enable --now cloud-netconfigd</code>
Validate	<code>cnctl status all</code> → <code>curl http://127.0.0.1:5209/api/status</code> → <code>journalctl -u cloud-netconfigd -n 30</code>
Config change	Edit YAML → <code>cnctl apply --dry-run</code> → copy to <code>cloud-network.yaml</code> → <code>systemctl restart cloud-netconfigd</code>
Reload signal	<code>cnctl reload</code> (prints systemd guidance) → <code>sudo systemctl reload cloud-netconfigd</code>
Upgrade binary	<code>systemctl stop cloud-netconfigd</code> → replace <code>/usr/bin/cloud-netconfigd</code> and <code>/usr/bin/cnctl</code> → <code>systemctl start cloud-netconfigd</code>
Multi-interface rollout	Start from <code>examples/multi-interface.yaml</code> → set <code>network.interfaces.enabled</code> → enable <code>network.routing.policy_routing</code>

cnctl command reference

Command	Purpose
cnctl status [system network all]	Daemon, cloud provider, and interface state (default: all)
cnctl apply [-c PATH] [--dry-run]	Validate (and optionally stage) configuration
cnctl reload [--force]	Reload guidance (systemd reload)
cnctl version	CLI and license version

Local HTTP API

Endpoint	Response
GET /health	OK
GET /api/status	JSON: status , provider , version
GET /api/cloud/status	OK

Bind address comes from `server.listen` (default `127.0.0.1:5209`).

Debug metadata or routing

1. Set `logging.level: debug` in config.
2. `sudo systemctl restart cloud-netconfigd` .
3. `journalctl -u cloud-netconfigd -f` .

Runtime state is written under `/run/cloud-network` (configurable via `state.directory`).

Operate (CLI)

1. Pick the workflow row that matches your job (install, validate, config change, upgrade).
2. Run the listed commands in order; use `--dry-run` before any restart that changes routing.
3. For mutating changes, schedule a maintenance window — policy routes affect egress paths.
4. **Empty / fail:** Capture `journalctl -u cloud-netconfigd` and `cnctl status network` before retrying.
5. **Success:** `cnctl status all` shows running daemon, expected provider, and addresses on supplementary NICs.